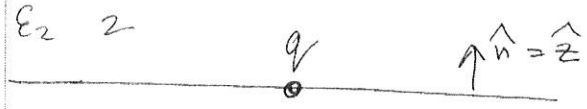
P1 Let charge q be at the origin.



Let $\vec{E} \sim q\frac{\vec{r}}{r^3} = q\frac{\hat{r}}{r^2}$ $\qquad \vec{E} = \begin{cases} A_1 \frac{q\vec{r}}{r^3}, & \text{for region 1 } (z<0) \\ A_2 \frac{q\vec{r}}{r^3}, & \text{for region 2 } (z>0) \end{cases}$

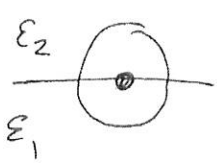
(1) This choice ensures boundary condition for the normal components $(\vec{D}_2 - \vec{D}_1)\cdot\hat{n} = \sigma_f = 0$ everywhere except for $\vec{r} = 0$ (where the charge is)

(2) Transverse components

$$(\vec{E}_2 - \vec{E}_1)\times\hat{n} = 0 \Rightarrow \frac{q}{r^3}(A_2 - A_1)\,\vec{r}\times\hat{z} = 0$$

$$\Rightarrow A_2 = A_1 = A.$$

Note that consideration in (1) avoids $\vec{r} = 0$ point where the charge is located. In terms of charge density $\rho_f = q\,\delta(\vec{r}) = q\,\delta(x)\,\delta(y)\,\delta(z)$ this is a very singular point.



However $\vec{\nabla}\cdot\vec{D} = \rho_f \Rightarrow \int \vec{\nabla}\cdot\vec{D}\,dV = \int \vec{D}\cdot d\vec{S} = \int \rho_f\,dV = q$

Because of the assumed spherical symmetry

$$\vec{D} = \vec{D}_1 = A\varepsilon_1 \frac{q\hat{r}}{r^2} \quad \text{for } z<0$$

$$\vec{D} = \vec{D}_2 = \varepsilon_2 A \frac{q\hat{r}}{r^2} \quad \text{for } z>0$$

$$\Rightarrow \int \vec{D}\cdot d\vec{S} = \varepsilon_1 A \frac{q\hat{r}}{r_0^2}\cdot 2\pi r_0^2 \hat{r} + \varepsilon_2 A \frac{q\hat{r}}{r_0^2}\cdot 2\pi r_0^2 \hat{r} = q$$

choose gaussian surface to be a sphere of radius r_0 $\Rightarrow$ area of a <u>hemisphere $= 2\pi r_0^2$</u>

$$\Rightarrow 2\pi A q(\varepsilon_1+\varepsilon_2) = q \Rightarrow A = \frac{1}{2\pi(\varepsilon_1+\varepsilon_2)}$$

$$\Rightarrow \vec{D} = \begin{cases} \dfrac{\varepsilon_1 q}{2\pi(\varepsilon_1+\varepsilon_2)} \dfrac{\hat{r}}{r^2} & z<0 \\[2ex] \dfrac{\varepsilon_2 q}{2\pi(\varepsilon_1+\varepsilon_2)} \dfrac{\hat{r}}{r^2} & z>0 \end{cases}$$

$$\Rightarrow \vec{E} = \frac{q}{2\pi(\varepsilon_1+\varepsilon_2)} \frac{\hat{r}}{r^2} \quad \text{for all } \vec{r}.$$

$$\vec{E} = -\vec{\nabla}\varphi \Rightarrow \varphi(\vec{r}) = \frac{q}{2\pi(\varepsilon_1+\varepsilon_2) r}$$

P2 $\varepsilon(\omega) = \varepsilon_0 \left(1 + \chi_e(\omega)\right)$, $\varepsilon'' = \varepsilon_0 \chi_e'' = \frac{\beta_0 \omega \tau}{1+\omega^2\tau^2}$.

Kramers-Kronig $\chi'(\omega) = \frac{2}{\pi} P\int_0^\infty \frac{x\,\chi''(x)}{x^2-\omega^2}\,dx$

$$\chi'(\omega) = \frac{2}{\pi}\frac{\beta_0}{\varepsilon_0} P\int_0^\infty dx \; x \frac{x\tau}{1+x^2\tau^2} \frac{1}{x^2-\omega^2} = \left(\begin{array}{l} y = x\tau \\ dx = \frac{dy}{\tau}\end{array}\right)$$

$$= \frac{2\beta_0}{\pi\varepsilon_0} P\int_0^\infty \frac{dy}{\tau} \frac{y^2}{\tau(1+y^2)} \frac{\tau^2}{y^2-\omega^2\tau^2} =$$

$$= \frac{2\beta_0}{\pi\varepsilon_0} P\int_0^\infty dy \frac{y^2}{1+y^2} \frac{1}{y^2-\omega^2\tau^2} = \left(\text{denote } \omega^2\tau^2 = u^2\right)$$

$$= \frac{2\beta_0}{\pi\varepsilon_0} P\int_0^\infty dy \left(\frac{1}{1+y^2} + \frac{u^2}{y^2-u^2}\right) \frac{1}{1+u^2}$$

$$\frac{y^2}{(1+y^2)(y^2-u^2)} = \left(\frac{1}{1+y^2} + \frac{u^2}{y^2-u^2}\right)\frac{1}{1+u^2} .$$

$$I_1 = P\int_0^\infty dy \frac{1}{1+y^2} = \frac{\pi}{2}$$ (Principal part "P" does not play a role since the integral is regular)

$$I_2 = P\int_0^\infty \frac{dy}{y^2-u^2} = (y = ux) = \frac{1}{u} P\int_0^\infty dx \frac{1}{x^2-1} = \frac{1}{u}\int_0^\infty dx \frac{1}{2}\left(\frac{1}{x-1} - \frac{1}{x+1}\right)$$

$$= \frac{1}{2u}\left\{\int_\varepsilon^{1-\varepsilon} \frac{dx}{x-1} + \int_{1+\varepsilon}^{L} \frac{dx}{x-1} - \int_0^L \frac{dx}{x+1}\right\}.$$ At the end we need to take limits: $\varepsilon \to +0$, $L \to \infty$

$$\int_\varepsilon^{1-\varepsilon} \frac{dx}{x-1} = (x = 1-z) = \int_1^\varepsilon \frac{-dz}{-z} = -\int_\varepsilon^1 d(\ln z) = \ln(\varepsilon)$$

$$\int_{1+\varepsilon}^{L} \frac{dx}{x-1} = (x = 1+z) = \int_{\varepsilon}^{L-1} \frac{dz}{z} = \ln \frac{L-1}{\varepsilon}.$$

$$\int_{0}^{L} \frac{dx}{x+1} = \ln\left(\frac{L+1}{1}\right) = \ln(L+1)$$

$$\Rightarrow I_2 = \frac{1}{2u}\left\{\ln(\varepsilon) + \ln\left(\frac{L-1}{\varepsilon}\right) - \ln(L+1)\right\} = (\varepsilon \text{ drops out!})$$

$$= \frac{1}{2u} \ln\left(\frac{L-1}{L+1}\right) \xrightarrow[L\to\infty]{} 0.$$

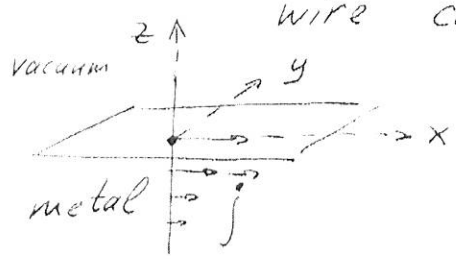
Thus $\chi'(\omega) = \frac{2\beta_0}{\pi\varepsilon_0} \frac{\pi}{2} \frac{1}{1+(\omega\tau)^2} = \frac{\beta_0}{\varepsilon_0(1+\omega^2\tau^2)}$

$$\chi(\omega) = \chi' + i\chi'' = \frac{\beta_0}{\varepsilon_0(1+\omega^2\tau^2)}(1+i\omega\tau) =$$

$$= \frac{\beta_0(1+i\omega\tau)}{\varepsilon_0(1+i\omega\tau)(1-i\omega\tau)} = \frac{\beta_0}{\varepsilon_0(1-i\omega\tau)}$$

$$\Rightarrow \varepsilon(\omega) = \varepsilon_0 \chi(\omega) = \frac{\beta_0}{1-i\omega\tau}.$$

Problem 3 When $\delta \ll a$ the surface of the wire can be approximated as flat



$\vec{j} = \sigma_0 \vec{E}$

both $\vec{j}$ and $\vec{E}$ are along x.

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} = i\omega \vec{B}$$

$$\nabla \times \vec{B} = \mu_0 \vec{j} = \mu_0 \sigma_0 \vec{E}$$ (displacement current is neglected as a small correction at $\omega \ll \omega_{pe}$)

$$\nabla \times (\nabla \times \vec{E}) = i\omega \mu_0 \sigma_0 \vec{E}$$

Since $\nabla \cdot \vec{E} = 0$ (why?) $\quad \nabla \times (\nabla \times \vec{E}) = -\nabla^2 \vec{E}$

$$\nabla^2 \vec{E} = -i\omega \mu_0 \sigma_0 \vec{E}$$

In our setup $\vec{E} = (E(z), 0, 0)$

$$\frac{d^2 E}{dz^2} = -i\omega \mu_0 \sigma_0 E$$ The solution, decaying when $z \to -\infty$ is

$$E(z) = E_0 e^{\sqrt{-i\omega\mu_0\sigma_0}\, z} = E_0 e^{\frac{z}{\delta}} e^{-\frac{iz}{\delta}}, \quad \delta = \sqrt{\frac{2}{\omega \mu_0 \sigma_0}}$$

as $\sqrt{-i} = \frac{-1-i}{\sqrt{2}}, \frac{1-i}{\sqrt{2}} \Rightarrow$ the second value should be chosen

We can now write actual electric field as the real part of the above solution:

$$E(z,t) = E_0 \, \mathrm{Re}\, e^{z/\delta} e^{-iz/\delta} e^{-i\omega t} =$$

$$= \rho_0 j_0 e^{z/\delta} \cos(\omega t + z/\delta), \quad \text{where } \rho_0 = \frac{1}{\sigma_0}$$

we introduced the amplitude of current j_0 at

the surface. It can be related to the total current via

$$I_0 = 2\pi a j_0 \int_{-\infty}^{0} dz\, e^{z/\delta} \cos\left(\frac{z}{\delta}\right) =$$

$$= 2\pi a j_0 \, \mathrm{Re} \int_{-\infty}^{0} dz\, e^{\frac{z}{\delta}(1-i)} =$$

$$= 2\pi a j_0 \, \mathrm{Re} \left(\frac{\delta}{1-i}\right) = \pi \delta a \quad \Rightarrow \boxed{j_0 = \frac{I_0}{\pi a \delta}}$$

Thus,

$$E(z,t) = \rho_0 \frac{I_0}{\pi a \delta} e^{z/\delta} \cos(\omega t + z/\delta)$$

Total Joule heat per unit length:

$$Q = 2\pi a \int_{-\infty}^{0} dz\, \overline{E(z,t) \cdot j(z,t)} = \frac{2\pi a}{\rho_0} \int_{-\infty}^{0} dz\, \overline{\left(E(z,t)\right)^2}$$

$$= \frac{2\pi a}{\rho_0} \left(\frac{\rho_0 I_0}{\delta \pi a}\right)^2 \int_{-\infty}^{0} dz\, e^{2z/\delta} \overline{\cos^2(\omega t + z/\delta)}$$

The time average is simply $\overline{\cos^2(\omega t + z/\delta)} = \frac{1}{2}$

and $\int_{-\infty}^{0} dz\, e^{2z/\delta} = \frac{\delta}{2}$, so that

$$\boxed{Q = \frac{\rho_0 I_0^2}{\pi a \delta} = \frac{I_0^2}{\pi a \delta \sigma_0}}$$

P4 <u>Ferrite</u>

$$\mu_{ij} = \begin{pmatrix} \mu_\perp & -i\mu_a & 0 \\ i\mu_a & \mu_\perp & 0 \\ 0 & 0 & \mu_\parallel \end{pmatrix} \qquad \vec{B} = \mu \vec{H} \Rightarrow B_i = \mu_{ij} H_j$$

$$\vec{D} = \varepsilon \vec{E}$$

Wave equation : $\vec{\nabla} \times \vec{H} = \frac{\partial \vec{D}}{\partial t} = \varepsilon \frac{\partial \vec{E}}{\partial t}$

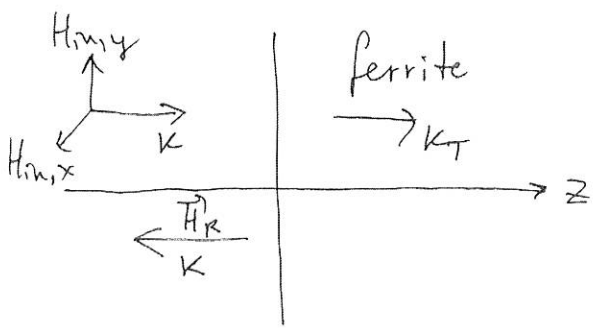
$$\underbrace{\vec{\nabla} \times (\vec{\nabla} \times \vec{H})}_{\vec{\nabla}(\vec{\nabla}\cdot\vec{H}) - \nabla^2 \vec{H}} = \varepsilon \frac{\partial}{\partial t} \underbrace{\vec{\nabla} \times \vec{E}}_{-\frac{\partial \vec{B}}{\partial t}} = -\varepsilon \frac{\partial^2}{\partial t^2} \vec{B}$$

Transverse wave : $\vec{\nabla} \cdot \vec{H} = 0$

$$\Rightarrow \frac{d^2}{dz^2} H_i = -\varepsilon \omega^2 B_i = -\varepsilon \omega^2 \mu_{ij} H_j$$

let $\vec{H}_T = (H_x, H_y, 0)\, e^{i k_T z}$ for $z > 0$.

$z < 0$ $\begin{cases} \text{Incident wave (Circularly polarized)} \quad \vec{H}_{in} = (H_{in}, i\nu H_{in}, 0)\, e^{ikz} \quad (\nu = \pm 1) \\ \text{Reflected} \quad \vec{H}_R = (H_{Rx}, H_{Ry}, 0)\, e^{-ikz} = H_R (1, i\nu, 0)\, e^{-ikz} \end{cases}$



$$\underbrace{k_T^2 H_j}_{k_T^2 \begin{pmatrix} H_x \\ H_y \end{pmatrix}} = \varepsilon \omega^2 \mu_{je} H_e = \varepsilon \omega^2 \begin{pmatrix} \mu_\perp & -i\mu_a \\ i\mu_a & \mu_T \end{pmatrix} \begin{pmatrix} H_x \\ H_y \end{pmatrix}$$

$$\begin{cases} (k_T^2 - \varepsilon\mu_\perp\omega^2)\, H_x = -i\mu_a \varepsilon\omega^2 H_y \\ (k_T^2 - \varepsilon\mu_\perp\omega^2)\, H_y = i\mu_a \varepsilon\omega^2 H_x \end{cases}$$

$$\Rightarrow \quad (k_T^2 - \varepsilon\mu_\perp\omega^2)^2 = (\mu_a \varepsilon\omega^2)^2$$

$$\Rightarrow \quad k_\pm^2 = k_T^2 = \varepsilon\mu_\perp\omega^2 \pm \mu_a\varepsilon\omega^2$$

For $k_+ = \sqrt{\varepsilon}\,\omega\sqrt{\mu_\perp + \mu_a}$: $\quad H_x = -i H_y$

$$\Rightarrow \quad \vec{H}_+ = H_+ (1, i, 0)\, e^{ik_+ z}$$

For $k_- = \sqrt{\varepsilon}\,\omega\sqrt{\mu_\perp - \mu_a}$: $\quad H_x = i H_y$

$$\Rightarrow \quad \vec{H}_- = H_- (1, -i, 0)\, e^{ik_- z}$$

eigenmodes are circularly polarized waves

<u>BC</u> $\quad H_{x,y}\big|_{z=-0} = H_{x,y}\big|_{z=+0} \Rightarrow (H_x \pm iH_y)\big|_{z=-0} = (H_x \pm iH_y)\big|_{z=+0}$

$$\Downarrow$$

circularly polarized wave basis can be used to describe continuity of $\vec{H}$.

$\Rightarrow$ if incoming wave is $\nu = +1$

$$\vec{H}_{in} = H_{in}(1, i, 0)\, e^{ikz}$$

then reflected and transmitted waves have the <u>same</u> circular polarization. Similarly for $\nu = -1$.

Thus $H_{in} + H_R = H_T$ (1)

Need another equ: continuity of $\vec{E}_{\parallel}$

$$-i\omega\varepsilon\vec{E} = \vec{\nabla}\times\vec{H} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial_x & \partial_y & \partial_z \\ H_x & H_y & 0 \end{vmatrix} = \hat{x}\left(-\partial_z H_y\right) + \hat{y}\,\partial_z H_x$$

since $H_{x,y} = H_{x,y}(z)$.

$\Rightarrow$ continuity of $E_{x,y}$ gives:

For $z=+0$: $\left.\dfrac{dH_y}{dz}\right|_{z=+0} = H_\pm(\pm i)(i k_\pm)$

$$\left.\frac{dH_x}{dz}\right|_{z=+0} = H_\pm(i k_\pm)$$

So, for $z<0$ $\vec{H} = \vec{H}_{in} + \vec{H}_R =$

$$= H_{in}(1, i\nu, 0)\,e^{ikz} + H_R(1, i\nu, 0)\,e^{-ikz}$$

$$\Rightarrow \left.\frac{d}{dz}H_x\right|_{z=-0} = H_{in}(ik) + H_R(-ik)$$

$$\left.\frac{d}{dz}H_y\right|_{z=-0} = H_{in}(i\nu)(ik) + H_R(i\nu)(-ik)$$

$$\frac{(ik)(H_{in}-H_R)}{-i\omega\varepsilon_0} = \frac{H_\pm(ik_\pm)}{-i\omega\varepsilon}$$

continuity of the y-component gives the same equation, just multiplied by (1-)

$$\Rightarrow \begin{cases} \frac{K}{\varepsilon_0}\left(H_m - H_R\right) = \frac{K_\pm}{\varepsilon} H_\pm & (2) \\ H_m + H_R = H_\pm & (1) \end{cases}$$

In order not to be confused by $(\pm)$ indices, let us consider $\nu = +1$ first:

$$\begin{cases} H_m + H_R = H_+ \\ \frac{K}{\varepsilon_0}\left(H_m - H_R\right) = \frac{K_+}{\varepsilon} H_+ = \frac{K_+}{\varepsilon}\left(H_{in} + H_R\right) \end{cases}$$

$$\Rightarrow \left(\frac{K}{\varepsilon_0} - \frac{K_+}{\varepsilon}\right) H_m = \left(\frac{K}{\varepsilon_0} + \frac{K_+}{\varepsilon}\right) H_R$$

$$H_R = \frac{\frac{K}{\varepsilon_0} - \frac{K_+}{\varepsilon}}{\frac{K}{\varepsilon_0} + \frac{K_+}{\varepsilon}} H_m$$

$$\Rightarrow H_T = H_+ = H_R + H_m = \frac{2\frac{K}{\varepsilon_0}}{\frac{K}{\varepsilon_0} + \frac{K_+}{\varepsilon}} H_m .$$

Energy flux: $\vec{S} = \frac{1}{2} \vec{E} \times \vec{H}^*$

$$\Rightarrow S_m = \frac{1}{2}\left(E_{in,x} H^*_{m,y} - E_{in,y} H^*_{m,x}\right)$$

$$E_{m,x} = \frac{1}{i\omega\varepsilon_0} \frac{dH_{in,y}}{dz} = \frac{iK\left(iH_m\right)}{i\omega\varepsilon_0} = \frac{iH_m K}{\omega\varepsilon_0}$$

(because $\left(H_m\right)_y = i H_m$).

$$E_{m,y} = -\frac{1}{i\omega\varepsilon_0} \frac{dH_{m,x}}{dz} = -\frac{iKH_m}{i\omega\varepsilon_0} = -\frac{K}{\omega\varepsilon_0} H_m$$

Therefore $S_m = \frac{1}{2}\left\{ \frac{i H_m K}{\omega \varepsilon_0} (i H_m)^* - \right.$

$$\left. - \left(- \frac{K H_m}{\omega \varepsilon_0} \right) H_m^* \right\} = \frac{K}{\omega \varepsilon_0} |H_m|^2$$

Obviously $S_R = \frac{-K}{\omega \varepsilon_0} |H_R|^2$

$\Rightarrow$ reflection coefficient is

$$R_+ = \frac{|S_R|}{|S_m|} = \left| \frac{H_R}{H_m} \right|^2 = \left(\frac{\frac{K}{\varepsilon_0} - \frac{K_+}{\varepsilon}}{\frac{K}{\varepsilon_0} + \frac{K_+}{\varepsilon}} \right)^2 .$$

$$K = \frac{\omega}{c}, \quad K_+ = \omega \sqrt{\varepsilon} \sqrt{\mu_\perp + \mu_a}$$

$$\frac{1}{c} = \sqrt{\varepsilon_0 \mu_0} \Rightarrow \frac{K}{\varepsilon_0} = \omega \frac{\sqrt{\varepsilon_0 \mu_0}}{\varepsilon_0} = \sqrt{\frac{\mu_0}{\varepsilon_0}}\, \omega$$

$$\frac{K_+}{\varepsilon} = \omega \sqrt{\frac{\mu_\perp + \mu_a}{\varepsilon}}$$

$$\Rightarrow R_+ = \left(\frac{\sqrt{\frac{\mu_0}{\varepsilon_0}} - \sqrt{\frac{\mu_\perp + \mu_a}{\varepsilon}}}{\sqrt{\frac{\mu_0}{\varepsilon_0}} + \sqrt{\frac{\mu_\perp + \mu_a}{\varepsilon}}} \right)^2$$ is expressed via impendance $Z = \sqrt{\frac{\mu}{\varepsilon}}$

Obviously, $$R_- = \left(\frac{\sqrt{\frac{\mu_0}{\varepsilon_0}} - \sqrt{\frac{\mu_\perp - \mu_a}{\varepsilon}}}{\sqrt{\frac{\mu_0}{\varepsilon_0}} + \sqrt{\frac{\mu_\perp - \mu_a}{\varepsilon}}} \right)^2 .$$

In terms of $\tilde{\varepsilon} = \frac{\varepsilon}{\varepsilon_0}$, $\tilde{\mu}_{\perp, a} = \frac{\mu_{\perp, a}}{\mu_0}$ can write

$$R_\pm = \left(\frac{1 - \sqrt{\frac{\tilde{\mu}_\perp \pm \tilde{\mu}_a}{\tilde{\varepsilon}}}}{1 + \sqrt{\tilde{\mu}_\perp \pm \tilde{\mu}}} \right)^2$$

Now, $R_\pm = 1$ when $\sqrt{\dfrac{\tilde{\mu}_\perp \pm \tilde{\mu}_a}{\tilde{\varepsilon}}} = \infty$ or 0.

$$\tilde{\mu}_\perp \pm \tilde{\mu}_a = 1 + \frac{\omega_M \omega_0}{\omega_0^2 - \omega^2} \pm \frac{\omega_M \omega}{\omega_0^2 - \omega^2} = 1 + \frac{\omega_M(\omega_0 \pm \omega)}{\omega_0^2 - \omega^2}$$

$$= 1 + \frac{\omega_M}{\omega_0 \mp \omega} = \frac{\omega_0 \mp \omega + \omega_M}{\omega_0 \mp \omega}$$

Therefore $\tilde{\mu}_\perp + \tilde{\mu}_a = \infty$ when $\omega = \omega_0$ (resonance)

since $\tilde{\mu}_\perp + \tilde{\mu}_a = 1 + \dfrac{\omega_M}{\omega_0 - \omega} = \dfrac{\omega_0 + \omega_M - \omega}{\omega_0 - \omega} \xrightarrow[\omega \to \omega_0]{} \infty$.

Observe that $\tilde{\mu}_\perp + \tilde{\mu}_a = 0$ for $\omega = \omega_0 + \omega_M$.

$$\Rightarrow R_+(\omega = \omega_0) = 1, \quad R_+(\omega = \omega_0 + \omega_M) = 1.$$

But $\tilde{\mu}_\perp - \tilde{\mu}_a = 1 + \dfrac{\omega_M}{\omega_0 + \omega} = \dfrac{\omega_0 + \omega_M + \omega}{\omega_0 + \omega}$

is always finite (and does not diverge) for physical $\omega > 0$.

Therefore reflection for $V = -1$ is never complete,

$R_-(\omega) < 1$ for all ω.